

The Modular Hopping Kernel on a 3+1D Diamond Waist Is the Casini–Huerta–Myers Envelope

Robert W. McGwier
CTO, Cohere Technology Group
GitHub: [n4hy](#)

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Abstract

A2 (Papers 21–23) said the modular Hamiltonian is local. Jacobson needs the local kernel to be the geometric modular Hamiltonian [1, 2]

$$K = 2\pi \int w T_{00}, \quad w = R^2 - r^2.$$

T_{00} contains the hop. The test is therefore on nearest-neighbour K_{ij} , not on the diagonal.

1d massless interval (the CHM theorem, used as an instrument gate): $\rho(K_{i,i+1}, w_{nn}) = -0.942$. 3+1D ball $R = 5$, $m = 0$: $\rho = -0.987$, residual $R_{\text{shape}} = 0.162$. At $mR = 0.5$: $\rho = -0.989$. A permutation of w collapses to $\rho = -0.076$. Auditor, different sizes: 1d $\rho = -0.984$; 3d $\rho = -0.987$. C2 PASS. [P] as a necessary-condition test on this field.

A first diagonal instrument failed the 1d theorem and was discarded. This is not $\eta = 1/4G$, not foam, and not de Sitter.

1 Gate

Locked on the hopping kernel. C0: 1d $\rho < -0.70$. C1: ball $\rho < -0.60$. C2: $R_{\text{shape}} < 0.50$. C3: permuted w has $|\rho| < 0.30$. C4: C1 at $mR = 0.5$. The sign is required: hops are -1 , so K_{ij} tracks $-w$.

2 Numbers

	$\rho(K_{nn}, w_{nn})$	R_{shape}	perm ρ
1d $L = 32$	-0.942	0.334	—
1d audit $L = 24$	-0.984	0.176	—
ball $R = 5$, $m = 0$	-0.987	0.162	-0.076
ball $R = 5$, $mR = 0.5$	-0.989	0.150	-0.077
ball audit $R = 4$	-0.987	0.161	$+0.054$

The hopping kernel tracks a CHM-*type* envelope on this lattice: bulk-peaked, radial, negative for hop -1 . An independent adversary confirmed C1/C2 on $N = 13, 15$. Catch: a linear $R - r$ weight can beat the quadratic on a small ball ($R = 3$). The parabola wins at $R = 4, 5$. The quadratic is not uniquely selected.

Together with the area law of Paper 24 this is the lattice input Jacobson 1995 uses to get Einstein plus a free Λ . The vacuum of that equation has $G_{\mu\nu} + \Lambda g_{\mu\nu} = 0$. That is the zero-Einstein-tensor step. It is not a selection of $\Lambda = 0$ or of de Sitter, and G is not measured.

If one *identifies* $S = \alpha A_{\text{cut}}$ with $A/(4G)$ then $G = a^2/(4\alpha) \approx 1.02 a^2$ using $\alpha = 0.245$. That is a cutoff identification, not a derivation.

Admission. *I took the step A2 still owed: the local kernel has the CHM shape. I threw out a diagonal instrument that failed its own 1d theorem. I did not write that G was measured, and I did not write that the foam is done.*

References

- [1] J.J. Bisognano and E.H. Wichmann, J. Math. Phys. 16 (1975) 985.
- [2] H. Casini, M. Huerta and R.C. Myers, JHEP 05 (2011) 036, arXiv:1102.0440.
- [3] T. Jacobson, Phys. Rev. Lett. 75 (1995) 1260.